

MEDTRONIC SINGLE CHAMBER 53401

ABOUT

The **Medtronic Single Chamber 53401 External Temporary Pacemaker** is designed for clinical environments requiring short-term atrial or ventricular pacing. Its intuitive interface and constant-current output ensure reliable therapy delivery, adapting to changes in lead impedance. With modes for both demand (synchronous) and asynchronous pacing, it supports therapeutic, prophylactic, and diagnostic applications. The device is MR Unsafe and must be used with continuous patient monitoring.

FEATURES

- **Multi-Mode Pacing:** Supports AAI, AOO, VVI, and VOO modes for flexible therapy.
- **Wide Rate Range:** Basic pacing (30–200 ppm) and rapid atrial pacing (80–800 ppm).
- **Adjustable Output:** 0.1–25 mA amplitude with a fixed 1.5 ms pulse width.
- **Enhanced Sensitivity:** 0.4–20 mV sensing to detect intrinsic rhythms.
- **Long Battery Life:** Up to 19 days with two AA alkaline batteries (IEC LR6).
- **Ergonomic Design:** Lightweight (499 g) and compact for one-handed operation.
- **Safety Features:** 200 ms blanking post-pace, 120 ms post-sense to prevent oversensing.



SPECIFICATIONS



DIMENSIONS

Height: 7.98 in (20.27 cm)

Width: 2.63 in (6.68 cm)

Depth: 1.63 in (4.14 cm)

Weight: 17.6 oz (499 g)

POWER

Battery Type: 2 × AA alkaline (LR6, e.g., Duracell MN1500).

Battery Life: Up to 19 days under typical use (Medtronic study, 2016).

OPERATIONAL

Pacing Modes: AAI, AOO, VVI, VOO.

Rate Ranges:

- Basic: 30–200 ppm.
- Rapid Atrial: 80–800 ppm.

Output: 0.1–25 mA (constant current).

Pulse Width: Fixed at 1.5 ms.

Sensitivity: 0.4–20 mV.

Blanking Periods:

- 200 ms $\pm$ 5/-30 ms (post-pace).
- 120 ms $\pm$ 2/-30 ms (post-sense).

Electrode Compatibility: Unipolar or bipolar leads.

ENVIRONMENTAL

- **Service Life:** 7 years.
- **MR Status:** MR Unsafe.